

# **Generation Interconnection Combined Feasibility / System Impact Study Report**

## **For Queue Project Q11-0076**

Bear 345 kV  
March 16, 2013

### ***1. Introduction***

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0076. The Interconnection Customer proposes a 591.2 MW natural gas combined cycle facility interconnecting at the Bear 345 kV substation in the Virginia area. The request was received on October 18, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 65 higher-queued active requests and their assigned network upgrades.

### ***2. Summary of Results***

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,465,000.

### ***3. Network Impacts***

#### ***3.1 Generator Deliverability and Single Contingency Analysis***

STATE ST-ROBISON PARK 230 KV: 129.1% loading (649 MVA) following the outage of the RICHLAND-DEKALB 500 KV; DFAX = 6.7%.

For the loss of DEKALB-JONES CHAPEL 500 KV, the JONES CHAPEL 500-230 kV Transformer loads to 123.7% of its 767 MVA rating. Project DFAX: 6.3%.

BLUEBONNET-RUSSELL FORK 230 KV: 109.9% loading (712 MVA) following the outage of the DEKALB-JONES CHAPEL 500 KV; DFAX = 6.2%.

For the loss of JNES CHAPEL-DEKALB 500 KV, the ROBISON PARK-RUSSELL FORK 230 KV L loads to 109.6% of its 919 MVA rating. Project DFAX: 8.5%.

For the loss of DEKALB-RICHLAND 500 KV line, the DEKALB 230-1 kV XFMR loads to 108.3% of its 869 MVA rating. Project DFAX: 7.2%.

The CHANCELLOR ROW-LEAVTT 345 KV line is overloaded to 104.4% of its 1,220 MVA emergency rating for the loss of the BIRCHOOD-LEAVITT SUBSTATION 345 KV L. The project contributes 22.5% DFAX (133.0 MW) to this overload.

DECLO-NEW LONDON 138 KV: 103.9% loading (257 MVA) following the outage of the FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV L; DFAX = 6.4%.

FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV: 103.6% loading (2,085 MVA) following the outage of the LEAVITT-BIRCHWOOD 345 KV Line; DFAX = 23.5%.

#### ***3.2 Multiple Facility Contingency Analysis***

No additional overloads were identified for multiple facility contingencies.

#### ***3.3 Short Circuit and Stability***

Short circuit and stability analyses identified no additional network upgrades.

#### 4. Required Network Upgrades and Cost Estimates

| Facility                                       | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| ROBISON PARK-STATE ST 230 KV                   | Rebuild line with higher-capacity conductor | \$12,843,000  | 13% (\$1,694,000)     |
| JONES CHAPEL 500-230 KV Transformer            | Replace transformer with higher-rated unit  | \$32,678,000  | 13% (\$4,358,000)     |
| BLUEBONNET-RUSSELL FORK 230 KV                 | Reconductor line                            | \$18,670,000  | 13% (\$2,477,000)     |
| ROBISON PARK-RUSSELL FORK 230 KV L             | Reconductor line                            | \$23,084,000  | 13% (\$3,061,000)     |
| DEKALB 230-1 kV XFMR                           | Replace transformer with higher-rated unit  | \$16,677,000  | 13% (\$2,219,000)     |
| CHANCELLOR ROW-LEAITT 345 KV                   | Reconductor line                            | \$10,895,000  | 28% (\$3,034,000)     |
| DECLO-NEW LONDON 138 KV                        | Reconductor line                            | \$14,634,000  | 16% (\$2,375,000)     |
| FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV | Reconductor line                            | \$36,578,000  | 17% (\$6,247,000)     |
|                                                | Total allocated to project                  |               | \$25,465,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.